FinInteract: Benchmarking Clarification and Intent Integration in Ambiguous Financial Question Answering

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Abstract

Large language model agents are increasingly deployed for financial information retrieval, and hence for downstream analytics and investment decisions. However, financial terminology is highly specific, and a question without sufficient specification may admit multiple defensible readings with materially different values, e.g., Meta Platforms' "operating income" is \$46.75B consolidated or \$62.87B for the Family of Apps segment, while each reading's ground truth is exact and programmatically verifiable. A capable agent should recognize the ambiguity and ask a clarification question instead of committing to a plausible but unintended reading. Existing financial benchmarks cannot see this, because they attach only one gold answer per question and therefore fail to distinguish agents that resolve the ambiguity from those making ungrounded guesses, a blind spot we call the *single-gold illusion*. To address this illusion, we release FinInteract, a bilingual (English/Chinese) benchmark of 173 instances that pairs each question with a *default* and an *intended* interpretation (fixed by a disambiguating context) across a five-category ambiguity taxonomy, and grades whether an agent elicits and then integrates the right clarification. Re-grading the *same* retrieval-augmented outputs against the intended rather than the default reading cuts GPT-4o's measured accuracy by a factor of 3.1 (37.0% to 12.1%), so a single-gold convention would have reported roughly three times the model's true competence. Our fine-grained analysis shows that this failure is one of interaction: models answer above 90% correctly once the interpretation is supplied, but reach at most 28.9% under their own clarification at full scale (44.0% in a cross-vendor pilot), so the bottleneck lies in issuing the correct clarification and then integrating the answer. We further report a skill metric that localizes failure modes of different financial ambiguities across two well-powered categories (entity scope and metric definition) and three exploratory ones.

1 Introduction

Large language model (LLM) agents are increasingly utilized in financial analysis to answer questions in regulatory filings (Islam et al. 2023; Chen et al. 2021; Reddy et al. 2024), and carry out multi-step research and retrieval over corporate disclosures (Bigeard et al. 2025; Choi et al. 2025; Krumdick et al. 2024; Xie et al. 2024). The agent reads a user's question, searches through relevant filings, and returns the answer for downstream actions.

However, financial questions are often deceptively underspecified. Financial terminology is highly specific, and a natural language question may map to several defensible readings for different numbers, e.g., in the question *"What was Meta Platforms' operating income?"* (Figure 1), an agent must decide whether the user means the consolidated GAAP figure (\$46.75B for FY2023) or the Family of Apps segment (\$62.87B), both being programmatically verifiable against SEC facts. A capable agent should recognize the ambiguity and follow up with the user instead of directly committing to a plausible but unintended reading.

Existing financial benchmarks cannot measure this as they grade only the final answer by assuming the question is disambiguated (Islam et al. 2023; Chen et al. 2021; Reddy et al. 2024; Krumdick et al. 2024; Xie et al. 2024). Without a grounding check, they cannot distinguish whether the agent actually resolves the ambiguity or guesses the common reading. We call this blind spot the *single-gold illusion* (Table 13 in Appendix D contrasts FinInteract with these benchmarks), which is currently underexplored in the financial domain relative to general ambiguous QA literature (Min et al. 2020; Li et al. 2025; Kuhn, Gal, and Farquhar 2023). We study financial ambiguity over three research questions:

- **RQ1**: Do frontier models recognize financial report ambiguity?
- **RQ2**: Which financial ambiguity categories lead to the most failures?
- **RQ3**: Can interventions at inference or training time improve resolution?

Answering these questions requires four design goals, which we state here and instantiate in Sec. 3. **(G1) Representative coverage**: the benchmark must draw real ambiguity from public filings. **(G2) Fine-grained diagnosis**: it must expose the distinct dimensions of ambiguity through paired default and intended readings, so that evaluation can localize which clarification ability a model lacks. **(G3) Low-cost, leak-safe construction**: it must be buildable semi-automatically, and the disambiguating context must never contain the answer. **(G4) High, human-validated quality**: ground truth must be programmatically verifiable and vali-

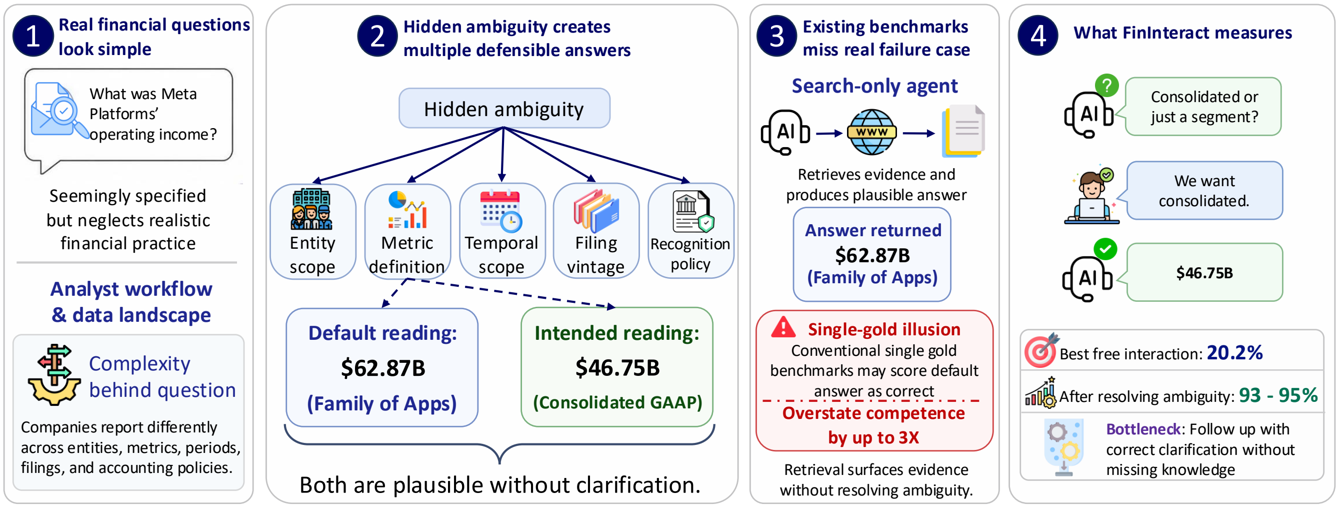

Figure 1: **Evaluating financial agents on ambiguous queries. (1–2)** A real query ("What was Meta Platforms' operating income?") looks fully specified but hides ambiguity across five categories (entity, metric, period, filing, policy), here giving two defensible answers: a default \$62.87B (Family of Apps segment) and the intended \$46.75B (consolidated GAAP). **(3)** A search-only agent returns the plausible default, which a single-gold benchmark scores as correct, the *single-gold illusion* that overstates competence by up to $3\times$. **(4)** FinInteract instead grades whether the agent *asks*: the GPT-5 agent reaches 20.2% under free interaction versus 93–95% once the ambiguity is resolved, so the gap lies in interaction, not missing knowledge.

dated by domain experts at minimal annotation cost.

We release FinInteract, a bilingual benchmark in English and Chinese containing 173 instances, classified into five financial ambiguity categories. Each question Q is paired with a *default* interpretation recording a possible non-expert assumption and an *intended* interpretation fixed by the disambiguating context. By grading the same model outputs against the default and intended interpretations, we effectively expose this single-gold illusion. FinInteract includes a skill metric that localizes which clarification ability the model lacks, and evaluates the agents under a ReAct protocol where the user simulator answers only from the disambiguating context (Figure 2) to measure the agent's quality of clarify-then-integrate action.

Our evaluation answers the research questions in turn. First, while frontier models detect ambiguity, they rarely resolve it effectively: they ask the correct clarification question but still **fail to convert it into the correct answer**, reaching at most 28.9% under their own clarification against a 93–95% ceiling once the interpretation is supplied (Sec. 4.2). Second, targeting is **uneven across ambiguity categories**: entity, metric, and temporal ambiguity are almost always named, while no model ever raises the recognition basis, though this last observation rests on an exploratory nine-instance category (Sec. 4.4). Third, **conditioning on the ambiguity category helps at both inference and training time**: a category-aware clarification policy lifts GPT-5 from 34.0% to 46.0% on a stratified pilot, and category-guided fine-tuning raises on-category targeting from 25 to 69 AC@1, suggesting that this benchmark is not only diagnostic but also actionable (Sec. 4.5 and Appendix C). This paper makes three contributions:

1. **Problem definition.** We formalize ambiguity in financial questions with five categories grounded in standard accounting and disclosure practice (Sec. 2).
2. **Benchmark.** We introduce FinInteract, a bilingual financial ambiguity dataset containing 173 instances, each paired with a default and an intended interpretation verified by domain experts (Sec. 3).
3. **Empirical findings.** Our evaluation shows that existing models commonly fail to return the correct final answer even when they have correctly clarified the intention, and that conditioning on the ambiguity category improves resolution at both inference and training time (Sec. 4).

2 Problem Formulation

We first define an ambiguous financial QA instance, then the five-category taxonomy, the entropy measure that scores instance difficulty, and the interaction protocol under which agents are evaluated. An instance (example in Table 1) consists of an ambiguous question Q over a company filing and a disambiguating context C, the minimal set of scope or definitional constraints that collapses Q into a unique intended answer A under the intended interpretation $\mathcal{I}_{\text{int}}$. The same question also admits a second correct answer $A_d \neq A$ under the default interpretation $\mathcal{I}_{\text{def}}$, the reading a non-expert would assume when seeing Q alone. Both $\mathcal{I}_{\text{int}}$ and $\mathcal{I}_{\text{def}}$ are structured records over the financial ambiguity taxonomy defined below.

Financial ambiguity taxonomy. We derive five categories of financial ambiguity from standard accounting and disclosure practice (Table 2). Each instance normally exercises one primary category, with one or two optional secondary categories when additional ambiguity is present.

Table 1: This example of an ambiguous financial question admits three plausible entity scopes, i.e., the consolidated total and two reportable segments, giving an entropy of 1.58 bits. Without the disambiguating context C, a model's default interpretation yields the wrong answer.

Field	Value
Q	What was Meta Platforms' operating income?
C	Total company (consolidated) results under U.S. GAAP for the year ended December 31, 2023.
A (intended)	\$46.75B *[consolidated GAAP]*
A_d (default)	\$62.87B *[Family of Apps segment]*
Ambiguity category	Entity scope
H_0	1.58 bits
Intended span	"Income from operations for 2023 was \$46.75 billion. . . "
Default span	"Family of Apps. . . Income from operations \$62,871. . . "

Disambiguation entropy. We quantify how ambiguous an instance is using the prior confidence entropy $H_0 = \log_2 k$, where k denotes the number of plausible interpretations of Q without clarifying interaction, e.g., the example in Table 1 has three plausible interpretations and hence an entropy of $H_0 = \log_2 3 \approx 1.58$ bits. H_0 also serves as our measure of difficulty and clarification efficiency in Sec. 3.5. Both A and A_d are defensible from publicly available filings, so an agent should be able to identify the ambiguity by reading Q alone without C. This pairing of default and intended interpretations generalizes the target-distractor design of InteractComp (Deng et al. 2026) to the financial domain.

Agent interaction model. We follow InteractComp's ReAct framework (Yao et al. 2023) where the agent may issue three types of actions, namely **search**(Q) to retrieve a passage from the filing corpus, **interact**(Q) to present a yes-or-no question to the simulated user who answers based on C, and **answer**(Q) to submit a final answer. Specifically, we evaluate agents in three primary modes (Table 3) and four ablation modes. We additionally include a category-conditioned interaction mode, described in Appendix C.7.

3 Methodology

FinInteract instantiates the formulation of Sec. 2 as a semi-automated ambiguous financial QA corpus built from public filings. Each instance carries a programmatically verifiable answer and a paired default and intended interpretation, produced by an LLM construction pipeline with minimal human annotation and validated by domain experts. Figure 2 shows a worked instance and how the benchmark scores an agent's two resolution paths.

3.1 Design goals and principles

Design goals. FinInteract targets four goals: **(G1) Representative coverage** to draw real ambiguity from public filings. **(G2) Fine-grained diagnosis** to contain diverse possible readings so that evaluation can localize which clarification ability a model lacks. **(G3) Low-cost, leak-safe construction** to build and verify the corpus at scale in a semi-automated setting. **(G4) High, human-validated quality** to validate the instances and judges by domain experts with minimal annotation cost.

Table 2: Five-category financial ambiguity taxonomy.

Category	Definition
Temporal scope	Ambiguity over which time period is intended: fiscal year ending September vs. calendar year; Q3 results vs. full year; TTM vs. point-in-time.
Metric definition	Ambiguity over accounting basis: GAAP net income vs. non-GAAP adjusted income; organic revenue vs. as-reported; operating EBITDA vs. net income.
Entity scope	Ambiguity over consolidation level: segment vs. consolidated total; parent-attributable vs. full consolidated; Class A vs. Class B shares; A-shares vs. H-shares.
Filing vintage	Ambiguity over filing version: original 10-K vs. amended 10-K/A; as-reported vs. restated; preliminary earnings release vs. final filing.
Recognition policy	Ambiguity over accounting policy: revenue recognized point-in-time vs. over time; gross vs. net revenue; capitalized vs. expensed R&D.

Table 3: We evaluate interaction modes including answer-only (A), answer with search (A+S) and answer with search and interaction (A+S+I).

Mode	Description
A	Pure parametric recall
A+S	Oracle retrieval augmented
A+S+I	Standard full interaction
Always-ask	≥ 1 follow-up question
Category-oracle	Provide primary category
Template-oracle	Fixed human-written question
Enumerate	Lists all interpretations and answers

Design principles. To satisfy the four goals, we build FinInteract under four principles. **(P1) Easy to verify, ambiguous to resolve**: ground truth is programmatically verifiable from SEC XBRL facts and CNINFO structured data, yet the question cannot be resolved correctly without the disambiguating context C. **(P2) No yes/no questions**: the agent must produce a grounded value rather than a binary judgment. **(P3) Company name included**: the entity itself is never ambiguous, using a capitalized proper noun in English and a CJK company name in Chinese. **(P4) Answer absent from** C: the agent cannot take the shortcut of copying directly from the context.

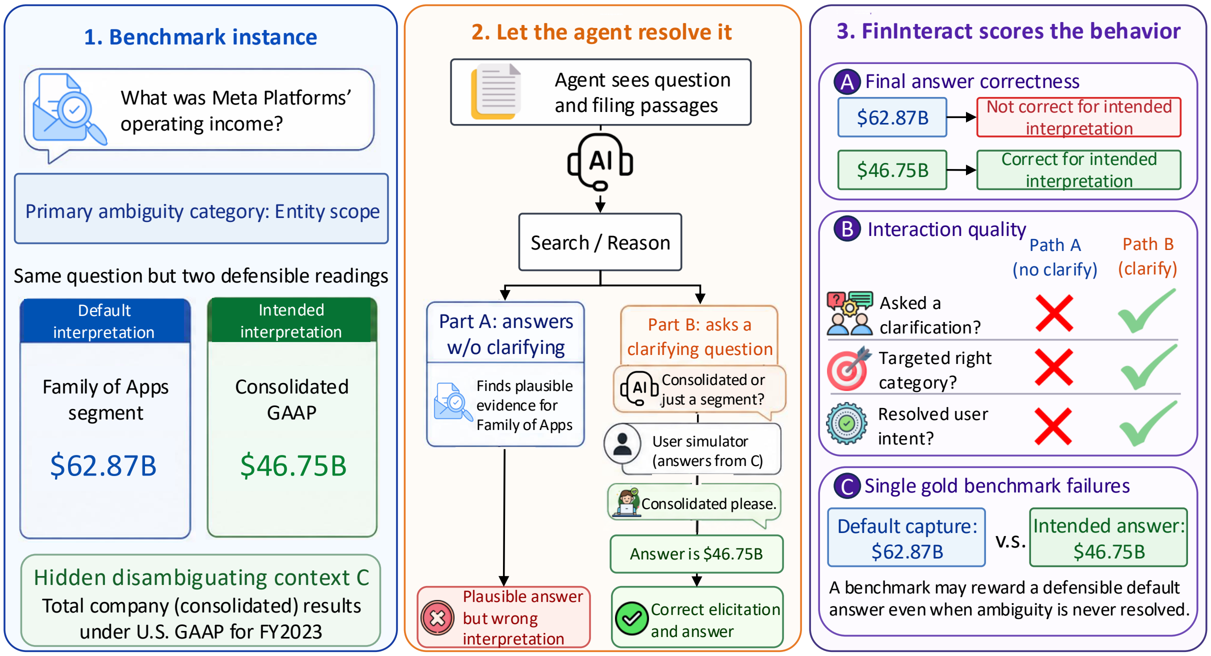

Figure 2: **FinInteract rewards actions that resolve the intended interpretation.** **(1)** Each instance carries a primary ambiguity category and two defensible readings, a default (Family of Apps, \$62.87B) and an intended (consolidated GAAP, \$46.75B), separated by a hidden disambiguating context C. **(2)** The agent either answers without clarifying (Part A, a plausible but wrong reading) or asks a question that a user simulator answers only from C (Part B, correct elicitation). **(3)** FinInteract scores three signals: final-answer correctness against the intended interpretation, interaction quality (did it ask, target the right category, and resolve intent), and the single-gold failure of rewarding the default.

3.2 Data sources

We build the instances from official filings. English passages, which make up about 80% of the candidate pool, come from SEC filings: DocFinQA long-context 10-K and 10-Q passages with verified QA pairs, and XBRL-derived facts for S&P 500 and Russell 1000 companies for fiscal years 2024–2025. Chinese passages come from the A-share annual reports of SSE 50 and CSI 300 constituents, retrieved through the CNINFO and akshare structured APIs. We avoid reusing published QA pairs directly to reduce memorization risk, though we do not claim absolute contamination safety. The current evaluation corpus contains 173 instances over 61 companies drawn from 2022–2025 filings, labeled with the taxonomy of Sec. 2; Table 4 gives the detailed breakdown. Specific details regarding passage counts and difficulty distribution are included in Appendix H.

Table 4: FinInteract dataset statistics. Disambiguation entropy $H_0 = \log_2 k$ over k plausible interpretations.

Property	Value	Property	Value
Total instances	173	**Category**: entity	94
Language: EN	53	metric	63
ZH	120	recognition	9
Source: EDGAR	50	temporal	7
CSRC-CNINFO	120	filing	0
DocFinQA	3	**# cat.**: single	128
Filing: 10-K	53	two	45
annual report	120	H_0: mean	2.20
Filing yr: 2022	40	median	2.00
2023	53	range	1.0–3.6
2024	58	**Blind-solve rate**	0.7%
2025	19	**Unique companies**	61

3.3 Curation

We curate each instance through a three-role pipeline. A **constructor** agent based on GPT-5 generates the full (Q, C, A) triple, together with the default and intended interpretations and the category label, from a filing passage; category-specific instructions prevent drift toward the categories that are easiest to generate. Next, a code-level **sanity check** rejects malformed candidates immediately, without API calls. Finally, an **adversarial verifier** runs ten trials that answer Q without the disambiguating context C, and rejects an instance if at least two of the ten produce the intended answer and the intended interpretation simultaneously. Details of the prompts, sanity-check rules, output schema, and cost are in Table 15, Appendix H, and Appendix G.

3.4 Validation

Two finance-literate annotators, holding the Chartered Financial Analyst (CFA) designation and a Hong Kong Securities and Futures Commission (SFC) license, independently review instances to check whether (1) the question Q is ambiguous, (2) the disambiguating context C uniquely identifies the intended reading without stating the answer, and (3) both the intended and the default answers are correct. A separate panel of five annotators supports the grader-validation and default-reading studies reported in Appendix B. We record substantial agreement, averaging 0.85 Gwet's AC1 (Gwet 2008), indicating strong chance-corrected consensus that, unlike Cohen's κ, stays reliable under the heavy skew toward "yes" labels. The full protocol and annotation details are in Appendix H.

3.5 Metrics

We adopt standard metrics from the ambiguous QA and interactive retrieval literature (Chen et al. 2021; Deng et al. 2026; Islam et al. 2023; Wei et al. 2025) so that FinInteract can be compared directly to prior benchmarks.

Accuracy (%) is the fraction of instances whose final answer matches the *intended* interpretation, graded by GPT-4o-mini under a financial-domain tolerance covering measurement differences, entity equivalence across stock markets, and currency normalization, with a fiscal-year mismatch counted as wrong.

Default-capture rate (%) is the fraction whose final answer instead matches the *default* interpretation. It measures how often a model returns the naive reading, letting us reconstruct what a conventional single-gold benchmark would have scored (Sec. 4.3).

Interaction diagnostics comprise the interaction rate (IR), the proportion of instances on which the agent issues at least one clarification question; the average number of clarification turns (Round); and the accuracy of targeting the true ambiguity category (AC@1). We report AC@1 in two forms, according to whether the question names any true category and whether it makes the single most important category its central concern.

Confidence is measured by the Expected Calibration Error (ECE) between stated confidence and correctness, capturing an agent's overconfidence on ambiguous queries.

Full metric definitions, including the accuracy grading tolerance, the AC@1 decomposition, and a supplementary clarification-efficiency score (DisE^{+}), are in Appendix A.

4 Experiments

4.1 Evaluation setup

Agent architecture. We evaluate agents under the ReAct (Yao et al. 2023) framework in the seven configurations of Table 3. The **interact** action presents a yes-or-no question to the simulated user, who must respond "yes", "no", or "I don't know".

Prompting strategy. FinInteract evaluates interactive agents under several interaction protocols: answer-only, answer with search, answer with search and interaction, and the category-conditioned policies. All models run under the same ReAct system prompt. Additional details are in Appendix C.7.

Models. We evaluate a diverse set of models. The proprietary set comprises the GPT family (GPT-5.6, GPT-5, GPT-4o, and GPT-5-mini), Claude-Sonnet-5, Gemini-3.5-flash, Grok-4.5, DeepSeek-V4-flash, GLM-5.2, and Kimi-K3. The open-source set comprises the dense Qwen3 family (4B, 8B, 14B, and 32B) and two mixture-of-experts models, Qwen3-30B-A3B and Qwen3.5-35B-A3B. Proprietary models are accessed through a unified OpenRouter harness.

Table 5: Forcing more clarification does not help accuracy, but *resolving* the ambiguity lifts every model to 93–95%. The gap is therefore the model's inability to elicit C for itself, not missing knowledge or grounding. Abbreviations follow Table 3.

Model	A	A+I	Forced(4)	**Oracle**
GPT-5	0.6	20.2	1.5	**95.4**
GPT-4o	0.6	4.6	0.0	**93.1**
GPT-5-mini	0.6	0.0	0.6	**95.4**
Qwen3-30B	1.2	11.6	–	**90.2**
Qwen3.5-35B	0.6	28.9	–	**91.9**

4.2 Overall performance

Even frontier models are vulnerable to financial report ambiguity (Table 6). No model exceeds 44% accuracy on the intended interpretation under free interaction, and the ceiling is low for proprietary and open-source models alike: the best full-scale model reaches 28.9% and the best cross-vendor pilot model 44.0%.

The opportunity to interact does not by itself improve accuracy. The interaction difference Δ (added-interaction minus plain-search accuracy) is non-positive for eleven of the sixteen evaluated models, so the failure is not one of asking but of converting a clarified intention into the correct answer. For the three proprietary models we run with full interaction diagnostics this reading is direct: they interact on 96–99% of instances and target a true ambiguity category at AC@1 0.86–0.94, so they do recognize which kind of ambiguity is present and do ask about it, yet still answer wrongly. The open models interact far less often (15–72%), so their negative Δ mixes under-asking with the same integration failure.

Two ablations bracket this gap (Table 5). Forcing four rounds of clarification *lowers* accuracy rather than raising it, which is consistent with agents getting lost in multi-turn conversations (Laban et al. 2026). Supplying the disambiguating context C as an oracle instead lifts every model to 93–95%. Because C never contains the final answer A, this improvement shows that the gap is not missing knowledge but the ability to resolve the interpretation and carry it into the answer. Appendix B reports the full analysis. **Finding 1: models can observe the ambiguity and answer correctly once the intention is made explicit, but fail to bridge the intermediate step from an updated intention to the correct answer.**

4.3 Failure of existing benchmarks

Because every instance carries both a default and an intended answer, we can re-grade the *identical* model outputs under the two conventions and read off what a conventional single-gold benchmark would have reported (Table 6(b)). In the retrieval-augmented regime where such benchmarks operate, grading against the default inflates the score sharply:

Table 6: **(a)** Main results, reporting accuracy (%) against the *intended* interpretation per mode, with Δ the +Interact minus +Search difference (full-scale rows are N=173 and the cross-vendor frontier rows, IR and ECE blank, are a preliminary N=50 pilot). **(b)** Re-grading the same outputs against the intended versus the default reading a conventional benchmark would treat as gold (Same abbreviation as Table 3). **(c)** Per-category targeting (AC@1), where recognition policy is a shared blind spot (= 0 for every model). Category sizes: entity (Ent.) 94, metric (Met.) 63, recognition (Rec.) 9, temporal (Tmp.) 7.

(a) Main results

Model	Acc (%) A	Acc (%) A+S	Answer+Search+Interact A+S	Δ	IR	AC@1	ECE
Proprietary models							
GPT-5.6	10.0	36.0	30.0	−6	–	.98	–
GPT-5	0.6	6.4	**20.2**	+13.8	98.8	.86	.53
GPT-4o	0.6	12.1	4.6	−7.5	99.4	.94	.82
GPT-5-mini	0.6	0.0	0.0	0.0	96.0	.94	.00
Claude-Sonnet-5	4.0	**46.0**	**44.0**	−2	–	.98	–
Grok-4.5	2.0	40.0	32.0	−8	–	.87	–
GLM-5.2	12.0	40.0	40.0	0	–	.96	–
Kimi-K3	16.0	38.0	34.0	−4	–	.98	–
DeepSeek-V4-flash	12.0	28.0	24.0	−4	–	.96	–
Gemini-3.5-flash	14.0	6.0	30.0	+24	–	.96	–
Open source models							
Qwen3-4B	0.0	8.1	12.1	+4.0	28	.85	–
Qwen3-8B	0.6	14.4	24.9	+10.5	15	.88	–
Qwen3-14B	0.0	18.5	22.5	+4.0	36	.74	–
Qwen3-32B	1.2	27.2	8.4	−18.8	61	.96	–
Qwen3-30B-A3B	1.2	28.3	11.6	−16.7	72	.90	–
Qwen3.5-35B-A3B	0.6	**35.8**	28.9	−6.9	36	.81	–
Human baseline (N=50)							
Human	–	–	**30.0**	–	100	.72	–

(b) Single-gold illusion

Model	Mode	Intended (true Acc)	Default (single-gold)	Δ
GPT-5	A	0.6	2.3	+1.7
	A+S	6.4	10.4	+4.0
	A+S+I	**20.2**	12.1	−8.1
GPT-4o	A	0.6	0.6	0.0
	A+S	12.1	**37.0**	**+24.9**
	A+S+I	4.6	6.4	+1.8
GPT-5-mini	A	0.6	0.0	−0.6
	A+S	0.0	0.0	0.0
	A+S+I	0.0	0.0	0.0

(c) Per-category targeting (AC@1)

Model	Ent.	Met.	Rec.	Tmp.
GPT-5	.89	.92	**.00**	1.00
GPT-4o	1.00	.97	**.00**	1.00
GPT-5-m	.99	1.00	**.00**	1.00
Q3-30B	.93	.96	**.00**	1.00
Q3.5-35B	.85	.87	**.00**	1.00

GPT-4o is credited 37.0% against the default but only 12.1% against the intended reading, so a single-gold benchmark would have reported $3.1\times$ its true competence. The inflation reverses only under interaction, where GPT-5 scores 20.2% intended against 12.1% default, which is precisely the point: asking is what converts a default-anchored guess into the intended answer, and it is the one behavior single-gold grading cannot see. Appendix B reports the full re-grading together with a human study confirming that annotators shown only Q and the passage record the default reading as gold 66% of the time. **Finding 2: a single-gold convention overstates competence by up to $3.1\times$, because one gold answer per question encodes the default reading rather than the intended one.**

4.4 Targeting varies by ambiguity category

Aggregate AC@1 hides where clarification actually succeeds, so we decompose targeting by ambiguity category (Table 6(c)). Models name the correct category almost always for entity scope, metric definition, and temporal scope, but never for recognition policy, where AC@1 is 0 for every model evaluated. A case study makes the missed distinction concrete. Asked "What were General Electric's revenues?", a model may answer \$67.95B or \$26.79B depending on the recognition basis: the former is GAAP revenue combining over-time service recognition with point-in-time product recognition, the latter counts only revenue recognized at a point in time. Models return a single figure without asking which basis is meant, defaulting to the assumption that "revenue" is well defined and passing over the ASC 606 distinction between point-in-time and over-time recognition. We read this as evidence that a domain-specific reporting practice is not reflected in current models' general clarification behavior, which is what motivates the category-conditioned interventions of Sec. 4.5.

This last observation carries an important caveat. Recognition policy is one of our three exploratory categories, with only n=9 instances, and an item-level audit (Appendix F) finds most of them to be ASC 606 timing disaggregations where neither value is the total a non-expert would assume. Our two human annotators also never targeted the category, so the blind spot is at least not model-specific. We therefore report it as a preliminary observation on a small sample rather than a quantitative result, and treat entity scope (n=94) and metric definition (n=63) as the two well-powered categories. **Finding 3: targeting accuracy is uneven across ambiguity categories; entity, metric, and temporal ambiguity are reliably named, while recognition policy is never raised, on preliminary evidence from a nine-instance category.**

4.5 Ambiguity category as training signal

The diagnosis so far is descriptive, so we ask whether the ambiguity category can also serve as a training signal. We run category-conditioned training on a Qwen3-4B policy,

Table 7: RLVR ladder on FinInteract with Qwen3-4B, held-out n=51. Category-guided supervision installs on-category clarification, with the leak-proof signals (AC@1 and interaction) rising almost entirely at the SFT stage. Accuracy is leak-confounded and is reported for completeness only.

Stage	AC@1	Interaction	Accuracy	Reward
Base	25.0	54.9	84.3	0.96
+ SFT	68.6	**100**	88.2	1.28
+ KTO	**70.6**	**100**	84.3	1.27
+ GRPO	66.7	**100**	88.2	**1.30**

supervised fine-tuning on category-guided demonstrations followed by reinforcement learning, teaching the model to interact and to target the correct category. On the held-out split the leak-proof signals rise almost entirely at the supervised stage, with AC@1 going from 25 to 69 and interaction from 55 to 100 (Table 7, Figure 3a), while the later stages move within run-to-run noise at n=51. Full multi-turn reinforcement learning confirms that the episode is trainable end to end, with clarifying turns rising from 5.7 to 8.6 (Figure 3b). At inference time the same idea, conditioning on the category, lifts GPT-5 from 34.0% to 46.0% on a stratified N=50 pilot (Appendix C.7). The gains concentrate on the data-abundant categories, so the signal is actionable but does not reach the recognition-policy blind spot of Sec. 4.4. Additional details are in Appendix C.7. **Finding 4: conditioning on the ambiguity category improves clarification at both training and inference time, so FinInteract provides an actionable signal and not only a diagnosis.**

5 Discussion

Ambiguity categories are different learning problems. Some clarification abilities are solved by scale, while others, in particular the recognition basis, resist it. A single aggregate interaction score therefore hides where the difficulty actually sits, and training curricula should budget data per category rather than uniformly.

Elicitation is a trainable skill that current models lack. Progress will come less from larger backbones, which barely move interaction accuracy, than from policies and objectives that reward asking about the right category at the right time. Our category-conditioned interventions are a first step in that direction, and both leave a large gap to the oracle ceiling.

Ambiguity persists in human-AI collaboration. Because models confidently return the default reading even when domain experts are in the loop, the safer behavior is often to ask rather than to answer. This motivates evaluation and interfaces that credit well-timed clarification and calibrated abstention instead of raw commitment.

The construction should transfer to further categories, languages, and domains. FinInteract is currently semi-automated. A benchmark of this form should ideally extend itself on categories whose evidence is abundant, and the paired default and intended construction should transfer to other languages, filing regimes, and high-stakes domains where a single gold answer hides genuine ambiguity.

We note the following limitations of this study.

1. **Scale.** A 173-instance corpus is small, a consequence of prioritizing per-instance construction rigor.
2. **Language balance.** The English-to-Chinese split is 53:120, which may favor models pre-trained more heavily in one of the two languages.
3. **Dominating categories.** Entity scope (n=94) and metric definition (n=63) dominate, against only n=9 for recognition policy, which suggests that current models are not able to generate instances of the rarer categories in the first place.
4. **Generator and evaluator bias.** Under API budget constraints we use the GPT-5 family for construction and simulation, which may bias results toward that family.
5. **Data contamination.** Instances are freshly constructed from FY2022–2025 facts, but we cannot rule out that recently released models were pre-trained on those filings.

Appendix F expands each of these.

6 Related Work

We include the most relevant works to FinInteract and defer additional financial and general-domain benchmarks as well as more widespread related work in Appendix D.

Financial QA benchmarks such as FinanceBench (Islam et al. 2023), FinQA (Chen et al. 2021), and DocFinQA (Reddy et al. 2024), together with recent agentic successors, enforce single-gold, single-turn evaluation and test numerical reasoning or retrieval without exploring whether a query is under-specified. To our knowledge, FinInteract is the first to make query ambiguity the central focus through paired default/intended interpretations and multi-turn interaction in the financial domain. We include comparison with existing financial QA benchmarks in Appendix D.

Ambiguous QA and clarifying questions. In the general domain, AmbigQA (Min et al. 2020), CLAMBER (Zhang et al. 2024), and QuestBench (Li, Kim, and Wang 2026) study whether a model identifies under-specification and asks the minimal clarifying question, building on a long clarifying-question line in retrieval and dialogue. InteractComp (Deng et al. 2026) is our direct methodological ancestor, which we extend to finance with a domain-specific ambiguity taxonomy, per-category targeting (AC@1), and programmatically verifiable XBRL-grounded evidence pairs.

Multi-turn interaction and selective answering. Large models often *lose track* over multi-turn conversations (Laban et al. 2026), and calibrated systems frequently do better by answering *selectively* or abstaining under ambiguity (Cole et al. 2023; Kuhn, Gal, and Farquhar 2023). Our findings echo this in the financial setting, where forcing extra clarification degrades accuracy, but FinInteract turns the observation into a controlled, per-category diagnosis of *which* clarification an agent should issue and *whether* issuing it resolves the query, instead of only measuring whether more turns help.

7 Conclusion

We introduced FinInteract, to our knowledge the first bilingual benchmark that evaluates financial search agents on gen-

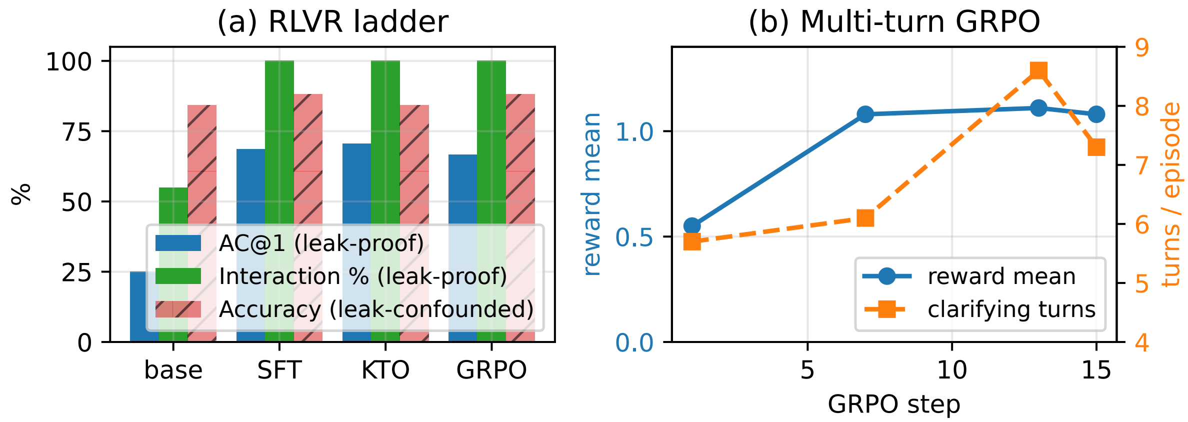

Figure 3: **Category-guided training installs on-category clarification.** (a) The leak-proof signals, AC@1 and interaction rate, jump at the supervised stage and change little thereafter. (b) True multi-turn GRPO on the 4B policy: mean reward and clarifying turns rise together over training.

uinely ambiguous queries. Its five-category ambiguity taxonomy, paired default and intended interpretation design, and construction pipeline with an adversarial verifier yield 173 instances that resist trivial disambiguation. Our central finding is that the bottleneck in handling financial QA ambiguity is interaction capability rather than financial knowledge, and that this capability has two parts: asking for the right clarification, and updating the answer once the clarification arrives. When the ambiguity is resolved for them, agents answer at 93–95% accuracy, against at most 28.9% when they must elicit the resolution themselves. This extends the interaction stagnation identified in InteractComp to a domain whose terminology is far more specific, and where the cost of a confident wrong answer is correspondingly higher. We further turn the diagnosis into an intervention in Sec. 4.5, showing that FinInteract also serves as an informative training signal, and we release it with full construction, evaluation, and training code for reproducible follow-on work.

References

Aliannejadi, M.; Kiseleva, J.; Chuklin, A.; Dalton, J.; and Burtsev, M. 2020. ConvAI3: Generating Clarifying Questions for Open-Domain Dialogue Systems (ClariQ). arXiv:2009.11352.

Bigeard, A.; Nashold, L.; Krishnan, R.; and Wu, S. 2025. Finance Agent Benchmark: Benchmarking LLMs on Real-world Financial Research Tasks. arXiv:2508.00828.

Chen, Y.; Wang, Y.; Yu, H.; Feng, T.; Zhu, S.; Zhang, M.; Patwary, M.; and You, J. 2026. Multi-Agent Evolve: LLM Self-Improve through Multi-Agent Co-evolution.

Chen, Z.; Chen, W.; Smiley, C.; Shah, S.; Borova, I.; Langdon, D.; Moussa, R.; Beane, M.; Huang, T.-H.; Routledge, B.; and Wang, W. Y. 2021. FinQA: A Dataset of Numerical Reasoning over Financial Data. In Moens, M.-F.; Huang, X.; Specia, L.; and Yih, S. W.-t., eds., *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 3697–3711. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics.

Choi, C.; Kwon, J.; Lopez-Lira, A.; Kim, C.; Kim, M.; Hwang, J.; Ha, J.; Choi, H.; Yun, S.; Kim, Y.; and Lee, Y. 2025. FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering. In *Proceedings of the 6th ACM International Conference on AI in Finance*, ICAIF '25, 632–637. New York, NY, USA: Association for Computing Machinery. ISBN 9798400722202.

Cole, J.; Zhang, M.; Gillick, D.; Eisenschlos, J.; Dhingra, B.; and Eisenstein, J. 2023. Selectively Answering Ambiguous Questions. In Bouamor, H.; Pino, J.; and Bali, K., eds., *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 530–543. Singapore: Association for Computational Linguistics.

Dalton, J.; Fischer, S.; Owoicho, P.; Radlinski, F.; Rossetto, F.; Trippas, J. R.; and Zamani, H. 2022. Conversational Information Seeking: Theory and Application. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR '22, 3455–3458. New York, NY, USA: Association for Computing Machinery. ISBN 9781450387323.

Deng, M.; Huang, L.; Fan, Y.; Kong, F.; Zhang, J.; Ren, F.; Bai, J.; Yang, F.; Miao, D.; Yu, Z.; Wu, Y.; Zhang, Y.; Teng, F.; Wan, Y.; Hu, S.; Li, Y.; Jin, X.; Hu, C.; Li, H.; Fu, Q.; Zhong, T.; Wang, X.; Tang, X.; Tang, N.; Wu, C.; and Luo, Y. 2026. InteractComp: Evaluating Search Agents With Ambiguous Queries. In *Forty-third International Conference on Machine Learning*.

Guo, X.; Xia, H.; Liu, Z.; Cao, H.; Yang, Z.; Liu, Z.; Wang, S.; Niu, J.; Wang, C.; Wang, Y.; Liang, X.; Huang, X.; Zhu, B.; Wei, Z.; Chen, Y.; Shen, W.; and Zhang, L. 2025. FinEval: A Chinese Financial Domain Knowledge Evaluation Benchmark for Large Language Models. In Chiruzzo, L.; Ritter, A.; and Wang, L., eds., *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies*

(Volume 1: Long Papers), 6258–6292. Albuquerque, New Mexico: Association for Computational Linguistics. ISBN 979-8-89176-189-6.

Gwet, K. L. 2008. Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1): 29–48.

Islam, P.; Kannappan, A.; Kiela, D.; Qian, R.; Scherrer, N.; and Vidgen, B. 2023. FinanceBench: A New Benchmark for Financial Question Answering. arXiv:2311.11944.

Krumdick, M.; Koncel-Kedziorski, R.; Lai, V. D.; Reddy, V.; Lovering, C.; and Tanner, C. 2024. BizBench: A Quantitative Reasoning Benchmark for Business and Finance. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 8309–8332. Bangkok, Thailand: Association for Computational Linguistics.

Kuhn, L.; Gal, Y.; and Farquhar, S. 2023. CLAM: Selective Clarification for Ambiguous Questions with Generative Language Models. arXiv:2212.07769.

Laban, P.; Hayashi, H.; Zhou, Y.; and Neville, J. 2026. LLMs Get Lost In Multi-Turn Conversation. In *The Fourteenth International Conference on Learning Representations*.

Li, B. Z.; Kim, B.; and Wang, Z. 2026. QuestBench: Can LLMs ask the right question to acquire information in reasoning tasks? In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Li, Z.; Li, Y.; Xie, H.; and Qin, S. J. 2025. CondAmbigQA: A Benchmark and Dataset for Conditional Ambiguous Question Answering. In Christodoulopoulos, C.; Chakraborty, T.; Rose, C.; and Peng, V., eds., *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2269–2288. Suzhou, China: Association for Computational Linguistics. ISBN 979-8-89176-332-6.

Mateega, S.; Georgescu, C.; and Tang, D. 2025. FinanceQA: A Benchmark for Evaluating Financial Analysis Capabilities of Large Language Models. arXiv:2501.18062.

Min, S.; Michael, J.; Hajishirzi, H.; and Zettlemoyer, L. 2020. AmbigQA: Answering Ambiguous Open-domain Questions. In Webber, B.; Cohn, T.; He, Y.; and Liu, Y., eds., *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 5783–5797. Online: Association for Computational Linguistics.

Rao, S.; and Daumé III, H. 2018. Learning to Ask Good Questions: Ranking Clarification Questions using Neural Expected Value of Perfect Information. In Gurevych, I.; and Miyao, Y., eds., *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2737–2746. Melbourne, Australia: Association for Computational Linguistics.

Reddy, V.; Koncel-Kedziorski, R.; Lai, V. D.; Krumdick, M.; Lovering, C.; and Tanner, C. 2024. DocFinQA: A Long-Context Financial Reasoning Dataset. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 445–458. Bangkok, Thailand: Association for Computational Linguistics.

Stelmakh, I.; Luan, Y.; Dhingra, B.; and Chang, M.-W. 2022. ASQA: Factoid Questions Meet Long-Form Answers. In Goldberg, Y.; Kozareva, Z.; and Zhang, Y., eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 8273–8288. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics.

Su, J.; and Cardie, C. 2026. Knowing but Not Showing: LLMs Recognize Ambiguity but Rarely Ask Clarifying Questions. arXiv:2605.25284.

Wei, J.; Sun, Z.; Papay, S.; McKinney, S.; Han, J.; Fulford, I.; Chung, H. W.; Passos, A. T.; Fedus, W.; and Glaese, A. 2025. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv:2504.12516.

Xie, Q.; Han, W.; Chen, Z.; Xiang, R.; Zhang, X.; He, Y.; Xiao, M.; Li, D.; Dai, Y.; Feng, D.; Xu, Y.; Kang, H.; Kuang, Z.; Yuan, C.; Yang, K.; Luo, Z.; Zhang, T.; Liu, Z.; Xiong, G.; Deng, Z.; Jiang, Y.; Yao, Z.; Li, H.; Yu, Y.; Hu, G.; Huang, J.; Liu, X.-Y.; Lopez-Lira, A.; Wang, B.; Lai, Y.; Wang, H.; Peng, M.; Ananiadou, S.; and Huang, J. 2024. FinBen: a holistic financial benchmark for large language models. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, NIPS '24. Red Hook, NY, USA: Curran Associates Inc. ISBN 9798331314385.

Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K. R.; and Cao, Y. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *The Eleventh International Conference on Learning Representations*.

Zhang, T.; Qin, P.; Deng, Y.; Huang, C.; Lei, W.; Liu, J.; Jin, D.; Liang, H.; and Chua, T.-S. 2024. CLAMBER: A Benchmark of Identifying and Clarifying Ambiguous Information Needs in Large Language Models. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 10746–10766. Bangkok, Thailand: Association for Computational Linguistics.